

PAST EXAMINATION QUESTIONS (FP2)

TOPIC 10: DIFFERENTIATION

1. A curve C has equation $\cos y = x$, for $-\pi < x < \pi$.

(i) Use implicit differentiation to show that

$$\frac{d^2y}{dx^2} = -\cot y \left(\frac{dy}{dx}\right)^2. \quad [4]$$

(ii) Hence find the exact value of $\frac{d^2y}{dx^2}$ at the point $\left(\frac{1}{2}, \frac{1}{3}\pi\right)$ on C . [2]

May/June 2019/11/12/1

2. The positive variables y and t are related by

$$y = a^t,$$

where a is a positive constant.

(i) (a) By differentiating $\ln y$ with respect to t , show that $\frac{dy}{dx} = a^t \ln a$. [3]

(b) Write down $\frac{d^2y}{dt^2}$. [1]

(ii) Determine the set of values of a for which the infinite series

$$y + \frac{dy}{dt} + \frac{d^2y}{dt^2} + \frac{d^3y}{dt^3} + \dots$$

[3]

is convergent.

A curve has parametric equations

$$x = t^a, \quad y = a^t.$$

(iii) Find $\frac{d^2y}{dx^2}$ in terms of a and t , and show that, when $t = 2$,

$$\frac{d^2y}{dx^2} = 2^{1-2a}(1 - a + 2 \ln a) \ln a. \quad [7]$$

May/June 2019/13/11E

3. The variables x and y are such that $y = -1$ when $x = 0$ and

$$\left(x + \frac{dy}{dx}\right)^3 = y^2 + x.$$

(i) Find the value of $\frac{dy}{dx}$ when $x = 0$. [1]

(ii) Find also the value of $\frac{d^2y}{dx^2}$ when $x = 0$. [4]

May/June 2018/13/1

4. The curve C has equation $2x^3 + 3x^2y - 3y^3 - 16 = 0$.

(i) Find the coordinates of the point A on C at which $\frac{dy}{dx} = 0$ and $x \neq 0$. $(2, -2)$ [5]

(ii) Find the value of $\frac{d^2y}{dx^2}$ at A . (0.5) [3]

Oct/Nov 2017/11/12/13/5

5. A curve C has equation $\tan y = x$, for $x > 0$.
(i) Use implicit differentiation to show that

$$\frac{d^2y}{dx^2} = -2x \left(\frac{dy}{dx} \right)^2. \quad [3]$$

- (ii) Hence find the value of $\frac{d^2y}{dx^2}$ at the point $\left(1, \frac{1}{4}\pi\right)$ on C . (**-0.5**) [2]

May/June 2017/11/12/3

6. A curve C has equation $x^3 - 3xy + y^2 = 4$. Find the value of $\frac{d^2y}{dx^2}$ at the point $(0,2)$ at C . $\left(\frac{9}{8}\right)$ [7]

May/June 2017/13/4

7. A curve C has equation $x^2 + 4xy - y^2 + 20 = 0$. Show that, at stationary points on C , $x = -2y$. [3]

Find coordinates of the stationary points on C , and determine their nature by considering their value of $\frac{d^2y}{dx^2}$ at the stationary points. (**max(4, -2), min(-4, 2)**) [8]

Oct/Nov 2016/11/12/13/8

8. The curve C is defined parametrically by

$$x = 2 \cos^3 t \quad \text{and} \quad y = 2 \sin^3 t, \quad \text{for } 0 < t < \frac{1}{2}\pi.$$

Show that, at the point with parameter t ,

$$\frac{d^2y}{dx^2} = \frac{1}{6} \sec^4 t \operatorname{cosec} t. \quad [4]$$

Oct/Nov 2015/11/12/13/1

9. A curve has equation $x^2 - 6xy + 25y^2 = 16$. Show that $\frac{dy}{dx} = 0$ at the point $(3,1)$. [4]

By finding the value of $\frac{d^2y}{dx^2}$ at the point $(3,1)$, determine the nature of this turning point. [5]
 $\left(y'' = -\frac{1}{16}; \text{max}\right)$

May/June 2015/11/12/6

10. The curve C has equation $x^2 + 2xy - 4y^2 + 20 = 0$. Show that if the tangent to C at the point (x, y) is parallel to the x -axis then $x + y = 0$. [3]

Hence find the coordinates of the stationary points on C , and determine their nature. [7]
(**max(2, -2); min(-2, 2)**)

May/June 2015/13/7

11. The curve C has parametric equation

$$x = \sin t, \quad y = \sin 2t, \quad \text{for } 0 \leq t \leq \pi.$$

Find $\frac{d^2y}{dx^2}$ in terms of t . $\left(\frac{d^2y}{dx^2} = \frac{4 \sin^3 t - 6 \sin t}{\cos^3 t}\right)$ [5]

Hence, or otherwise, find the coordinates of the stationary points on C and determine their nature. $\left(\max\left(\frac{1}{\sqrt{2}}, 1\right), \min\left(\frac{1}{\sqrt{2}}, -1\right)\right)$ [5]

May/June 2014/13/7

12. A curve has parametric equations

$$x = 2\theta - \sin 2\theta, \quad y = 1 - \cos 2\theta, \quad \text{for } -3\pi \leq \theta \leq 3\pi.$$

Show that

$$\frac{dy}{dx} = \cot \theta,$$

except for certain values of θ , which would be stated. ($\theta \neq k\pi, k$ is an integer) [4]

Find the value of $\frac{d^2y}{dx^2}$ when $\theta = \frac{1}{4}\pi$. (-1) [3]

Oct/Nov 2013/13/4

13. Show that $\frac{dy}{dx} = -\frac{4}{3}$ at the point $A(1, -2)$ on the curve with equation

$$y^3 - 3x^2y + 2 = 0,$$

and find the value of $\frac{d^2y}{dx^2}$ at A . $\left(-\frac{20}{27}\right)$ [8]

May/June 2013/13/4

14. (i) Use the definitions $\tanh \theta = \frac{e^\theta - e^{-\theta}}{e^\theta + e^{-\theta}}$ and $\operatorname{sech} \theta = \frac{2}{e^\theta + e^{-\theta}}$ to prove the results
(b) $\frac{d}{d\theta} (\tanh \theta) = \operatorname{sech}^2 \theta$. [4]

CAIE 1348/May/June 2013/1/13 (part)

15. The curve C has equation $x^3 + y^3 = 3xy$, for $x > 0$ and $y > 0$. Find a relationship between x and y when $\frac{dy}{dx} = 0$. ($y = x^2$) [4]

Find the exact coordinates of the turning point of C , and determine the nature of this turning point. $\left(\max\left(2^{\frac{1}{3}}, 2^{\frac{2}{3}}\right)\right)$ [8]

Oct/Nov 2012/13/10

17. The curve C has equation

$$xy + (x + y)^3 = 1.$$

Show that $\frac{dy}{dx} = -\frac{3}{4}$ at the point $A(1, 0)$ on C . [3]

Find the value of $\frac{d^2y}{dx^2}$ at $A. \left(\frac{9}{32}\right)$ [5]

May/June 2012/13/3

18. The point $P(2,1)$ lies on the curve with equation

$$x^3 - 2y^3 = 3xy.$$

Find

(i) The value of $\frac{dy}{dx}$ at $P. \left(\frac{3}{4}\right)$ [3]

(ii) The value of $\frac{d^2y}{dx^2}$ at $P. \left(\frac{1}{16}\right)$ [4]

Oct/Nov 2011/11/12/5

19. A curve has parametric equations

$$x = 2 \sin 2t, \quad y = 3 \cos 2t,$$

for $0 < t < \frac{1}{2}\pi$. For the point on the curve where $t = \frac{1}{3}\pi$, find the value of

(i) $\frac{dy}{dx}, \left(\frac{3\sqrt{3}}{2}\right)$ [3]

(ii) $\frac{d^2y}{dx^2} \cdot (6)$ [4]

Oct/Nov 2011/13/4

20. The curve C has equation

$$2xy^2 + 3x^2y = 1.$$

Show that, at the point $A(-1,1)$ on C , $\frac{dy}{dx} = -4$. [3]

Find the value of $\frac{d^2y}{dx^2}$ at $A. (-42)$ [5]

May/June 2011/13/4

21. The variables x and y are such that $y = -1$ when $x = 1$ and

$$x^2 + y^2 + \left(\frac{dy}{dx}\right)^3 = 29.$$

Find the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when $x = 1. \left(3; \frac{4}{27}\right)$ [5]

May/June 2010/11/12/1

22. It is given that

$$x = t^2 e^{-t^2} \quad \text{and} \quad y = t e^{-t^2}.$$

(i) Show that

[3]

$$\frac{dy}{dx} = \frac{1 - 2t^2}{2t - 2t^3}.$$

(ii) Find $\frac{d^2y}{dx^2}$ in terms of t . [5]

$$\left(\frac{(-1 + t^2 - 2t^4)e^{t^2}}{4t^3(1 - t^2)^3} \right)$$

May/June 2010/13/7

23. It is given that

$$x = t + \sin t, \quad y = t^2 + 2 \cos t,$$

where $-\pi < t < \pi$. Find $\frac{dy}{dx}$ in terms of t . $\left(\frac{dy}{dx} = \frac{2t - 2 \sin t}{1 + \cos t} \right)$ [2]

Show that

$$\frac{d^2y}{dx^2} = \frac{2t \sin t}{(1 + \cos t)^3}.$$
 [4]

Show that $\frac{dy}{dx}$ increases with x over the given interval of t . $\left(\frac{d^2y}{dx^2} > 0 \right)$ [2]

Oct/Nov 2009/1/4

24. A curve has equation

$$(x + y)(x^2 + y^2) = 1.$$

Find the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at the point $(0,1)$. $\left(-\frac{1}{3}, -\frac{4}{9} \right)$ [7]

May/June 2009/1/6

25. The curve C has equation

$$x^2 - xy - 2y^2 = 4.$$

Show that, at the point $A(2,0)$ on C , $\frac{dy}{dx} = 2$. [2]

Find the value of $\frac{d^2y}{dx^2}$ at A . (-9) [5]

Oct/Nov 2008/1/5

26. The positive variables x and y are related by

$$y = x^2 + 2 \ln(xy).$$

Find the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when both x and y are equal to 1. $(-4; 32)$ [7]

Oct/Nov 2007/1/5

27. Given that the variables x and y are related by

$$y = x + e^{-xy},$$

find the value of $\frac{d^2y}{dx^2}$ when $x = 0$. (1) [6]

May/June 2007/1/4

28. The curve C has equation

$$y = x^2 + \lambda \sin(x + y),$$

where λ is a constant, and passes through the point $A\left(\frac{1}{4}\pi, \frac{1}{4}\pi\right)$. Show that C has no tangent which is parallel to the y -axis. [5]

Show that, at A ,

$$\frac{d^2y}{dx^2} = 2 - \frac{1}{64}\pi(4 - \pi)(\pi + 2)^2. [5]$$

Oct/Nov 2006/1/10

29. Given that

$$x = t - \sin t \quad \text{and} \quad y = 1 - \cos t,$$

where $0 < t < 2\pi$, show that

$$\frac{d^2y}{dx^2} = -\frac{1}{4} \operatorname{cosec}^4\left(\frac{1}{2}t\right) [5]$$

May/June 2005/1/2

30. The curve C has equation

$$xy + (x + y)^5 = 1.$$

(i) Show that $\frac{dy}{dx} = -\frac{5}{6}$ at the point $A(1,0)$ on C . [3]

(ii) Find the value of $\frac{d^2y}{dx^2}$ at A . $\left(\frac{5}{27}\right)$ [5]

Oct/Nov 2004/1/7

31. The variables x and y are related by the equation $x^4 + y^4 = 1$, where $0 < x < 1$ and $0 < y < 1$.

(i) Obtain an equation which relates $x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}$, and deduce that

$$\frac{d^2y}{dx^2} = -\frac{3x^2}{y^7}. [6]$$

May/June 2003/1/7 (part)

32. A curve has equation $x^3 + xy^2 - y^3 = 3$.

(i) Show that there is no point on the curve at which $\frac{dy}{dx} = 0$. [4]

(ii) Find the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at the point $(1, -1)$. $\left(\frac{dy}{dx} = \frac{4}{5}, \frac{d^2y}{dx^2} = \frac{198}{125}\right)$ [5]

Oct/Nov 2002/1/6

33. Given that

$$x = 1 + \frac{1}{t} \quad \text{and} \quad y = t^3 e^{-t},$$

where $t \neq 0$, find $\frac{dy}{dx}$ in terms of t . $\left(\frac{dy}{dx} = t^4(t-3)e^{-t}\right)$ [3]

Find $\frac{d^2y}{dx^2}$ in terms of t , and hence find the values of t for which $\frac{d^2y}{dx^2} = 0$. $\left(\frac{d^2y}{dx^2} = e^{-t}t^5(t-2)(t-6),; t = 2, 6\right)$ [5]

May/June 2002/1/7

34. (a) It is given that $y = \operatorname{cosech}^{-1}x$. Show that for $x > 0$,

$$\frac{dy}{dx} = -\frac{1}{x\sqrt{x^2+1}},$$

and write down an expression for $\frac{dy}{dx}$ in terms of x for $x < 0$. $\left(-\frac{1}{x\sqrt{x^2+1}}\right)$ [4]

Nov 1995/1/3 (part)

35. (a) The real variables x, y and t are related by $x = \tanh t$ and $y = t^2 + 1$. Find $\frac{d^2y}{dx^2}$ in terms of t and hence show that, for all t ,

$$\frac{d^2y}{dx^2} \leq 2(2|t| + 1) \cosh^4 t. \quad [7]$$

$$\left(\frac{d^2y}{dx^2} = 2 \cosh^3 t (\cosh t + 2t \sinh t)\right)$$

Nov 1994/1/3 (part)

36. (b) Given that the equation of a curve is

$$y = x + \tanh^{-1}(x + y),$$

Show that, at all points of the curve except the origin,
(i) $\frac{dy}{dx} = 1 - \frac{2}{(x+y)^2}$, [3]

(ii) $\frac{d^2y}{dx^2} = 8 \cosh^3(x-y) \operatorname{cosech}^5(x-y)$. [5]

June 1993/1/3 (part)

37. Find the derivative with respect to x of

(ii) $\sin^{-1}\left(\frac{x}{10}\right) \cdot \left(\frac{1}{\sqrt{100-x^2}}\right)$

Nov 1991/1/7 (part)

38. (a) Given that $x = \cosh^3 t$ and $y = \sinh^3 t$, where $t \in \mathbb{R}$, show that $\frac{dy}{dx} = \frac{3}{5}$ when $t = \ln 2$.

Find the exact value of $\frac{d^2y}{dx^2}$ when $t = \ln 2$, giving your answer as a fraction in its lowest terms. $\left(\frac{1024}{5625}\right)$ [7]

June 1989/1/3 (part)

39. Differentiate with respect to x :

(ii) $\sin^{-1}(x^2)$.

June 1986/1/8 (part)

40. (a) For the curve which is defined parametrically by $x = \sinh^{-1} t$ and $y = \tanh^{-1} t$, where $-1 < t < 1$, express $\frac{dy}{dx}$ in terms of t . Hence show that

$$\frac{d^2y}{dx^2} = \frac{3t + t^3}{(1 - t^2)^2}.$$

[7]

Draw a sketch of this curve.

June 1987/1/3 (part)

41. [In this question, $\sin^{-1} u$ denotes the principal value.]

Let $y = \sin^{-1}(\sqrt{x})$, where $0 \leq x \leq 1$. Express $\frac{dy}{dx}$ in terms of x , and show that $\frac{dy}{dx} \geq 1$ for $0 < x < 1$. $\left(\frac{1}{2\sqrt{x-x^2}}\right)$ [5]

Sketch the graph of y .

By considering your sketch, show that

$$\int_0^1 \sin^{-1}(\sqrt{x}) dx + \int_0^{\frac{1}{2}\pi} \sin^2 y dy = \frac{\pi}{2},$$

[3]

and hence or otherwise evaluate $\int_0^1 \sin^{-1}(\sqrt{x}) dx$. $\left(\frac{\pi}{4}\right)$ [5]

June 1982/2/20

42. The curve C is defined, for all x , by the equation $y = \frac{x}{\cosh x}$.

(a) Show that at a stationary point of C , $x \tanh x = 1$.

(b) By drawing sketch graphs of $y = \tanh x$ and $y = \frac{1}{x}$, or otherwise, show that C has exactly two stationary points.

(c) Show that at either stationary point $\frac{d^2y}{dx^2} = -y$.

(d) Draw a sketch graph of C .

[You may assume that $y \rightarrow 0$ as $x \rightarrow \pm\infty$.]

MB November 1984/1/21

43. (Throughout this question, $-\frac{1}{2}\pi \leq \sin^{-1} x \leq \frac{1}{2}\pi$)

(i) Prove that, for $-1 < x < 1$, $\frac{d}{dx}(\sin^{-1} x) = (1 - x^2)^{-1/2}$.

June 1982/2/20 (part)

44. (a) (i) Prove that $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$, where $|x| < 1$.

(ii) Given that $y = \sin^{-1}[2x/\sqrt{1-x^2}]$, show that $\frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$.

June 1979/2/20 (part)

45. If $y = \tanh x$, show that $\frac{dy}{dx} = 1 - y^2$.

Hence, or otherwise, express $\frac{d^2y}{dx^2}$ and $\frac{d^3y}{dx^3}$ in terms of y . $\left(-\frac{2y}{1-y^2}; (6y^2 - 2)(1 - y^2)^2\right)$

MB June 1976/2/20

46. (c) If $u = e^x \cosh x$ and $v = e^x \sinh x$, show that $\frac{du}{dx} = \frac{dv}{dx}$ and give a relation between u and v which is independent of x . (**$u = v + 1$**)

MB November 1975/2/22 (part)

47. (b) If $y = \sinh^{-1}\sqrt{x^2 - 1}$, where $x > 1$, express x in terms of y . Hence, or otherwise, express $\frac{dy}{dx}$.

(i) in terms of x only, $\left(\frac{1}{\sqrt{x^2-1}}\right)$

(ii) in terms of y only. (**$\operatorname{cosech} y$**)

MB June 1975/2/21 (part)

48. (a) Differentiate, with respect to x , $\tanh^{-1}(\sin x)$ and $\sin^{-1}(\tanh x)$, simplifying your answer in each case. (**$\sec x$; $\operatorname{sech} x$**)

MB November 1974/1/22 (part)